

Prepared by: Abrar Ahmed

IoT Networking

Q.1 In which terms is the Internet's network architecture expressed?	
A. Networks used	B. Model used for node connections
C. Hardware links used	D. Internet Protocol Suite
Correct Answer: D	
Reference: Book name: IoT Networking, Page no 17	

Q.2 How many physical channels are required to link 'n' devices in a mesh topology?	
A. $n(n-1)/2$	B. $(n-1)/2$
C. $n/2$	D. $n(n-1)$
Correct Answer: A	
Reference: Book name: IoT Networking, Page no 14	

Q.3 Which of the following is not an advantage of the client/server network architecture?	
A. Enhanced network performance	B. High memory server requirements
C. Increased resource sharing speed	D. Improved security
Correct Answer: B	
Reference: Book name: IoT Networking, Page no 19	

Q.4 What is the advantage of using client/server network architecture over P2P network architecture?	
A. Easy configuration and maintenance	B. Minimum node failure impact
C. Economical	D. Backup
Correct Answer: D	
Reference: Book name: IoT Networking, Direct reference not available	

Q.5 Which of the following layers can have multiple session addresses?	
A. Transport layer	B. Network Layer
C. Data Link Layer	D. Session Layer
Correct Answer: A	
Reference: Book name: IoT Networking, Page no 31	

Q.6 Which network topology has a signal central hub connected to devices through cables?	
A. Bus	B. Star
C. Mesh	D. Ring
Correct Answer: B	
Reference: Book name: IoT Networking, Page no 13	

Q.7 Which organization allots Internet resources such as addresses?	
A. Internet Security	B. Internet Architecture
C. Internet Engineering Task Force	D. ISOC
Correct Answer: D	
Reference: Book name: IoT Networking, Page no 11	

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Q.8 Which of the following is a framework used to specify network physical components, configuration and functional organization?	
A. Network Topology	B. Internet
C. Web	D. Network Architecture
Correct Answer: D	
Reference: Book name: IoT Networking, Page no 16	

Q.9 Which of the following is a disadvantage of P2P network architecture?	
A. Configuration and Maintenance	B. No impact due to node failure
C. Security	D. Economical
Correct Answer: C	
Reference: Book name: IoT Networking, Page no 18	

Q.10 Which network topology does not support bi-directional data transfer?	
A. Bus	B. Star
C. Mesh	D. Ring
Correct Answer: A	
Reference: Book name: IoT Networking, Page no 12	

Q.11 Which of the following has a core at the center of the cable surrounded by cladding, buffer and jacket?	
A. Fiber optic cable	B. UTP
C. Co-axial cable	D. STP
Correct Answer: A	
Reference: Book name: IoT Networking, Page no 27	

Q.12 How many nodes can twisted-pair cables host to the maximum?	
A. 32	B. 1024
C. 64	D. 1064
Correct Answer: B	
Reference: Book name: IoT Networking, Page no 26	

Q.13 Which of the following is a device used to split or share a network connection among multiple computers?	
A. Router	B. NIC card
C. Hub	D. Switch
Correct Answer: C	
Reference: Book name: IoT Networking, Page no 28	

Q.14 Which of the following is not required for linking hypertext to internet?	
A. HTML	B. HTTP
C. Devices	D. URL
Correct Answer: C	
Reference: Book name: IoT Networking, Page no 10	

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Q.15 Which network topology has a monitor node or station for performing all functions of the network?	
A. Bus	B. Star
C. Mesh	D. Ring
Correct Answer: D	
Reference: Book name: IoT Networking, Page no 13	

Q.17 Which of the following architecture is turned as intelligent network?	
A. P2P	B. Hybrid
C. Telephone network	D. Internet
Correct Answer: C	
Reference: Book name: IoT Networking, Page no 17	

Q.18 Which of the following devices is used to share a single Internet connection with multiple networks?	
A. Switch	B. LAN Card
C. Router	D. Hub
Correct Answer: C	
Reference: Book name: IoT Networking, Page no 28	

Q.19 What does an OSI model specify for a communication system?	
A. Rules	B. Processes
C. Protocols	D. Services
Correct Answer: B	
Reference: Book name: IoT Networking, Page no 30	

Q.20 Which of the following is not an advantage of RAD?	
A. Reduced development risk	B. Time-to-market
C. Fixed design	D. Flexibility for new features
Correct Answer: D	
Reference: Book name: IoT Networking, Page no 43	

Q21: How many set of group does numerical representation of Ipv6 address have?	
A. eight	B. six
C. four	D. ten
Correct Answer: A	
Reference: Book name IoT NETWORKING, page no 32	

Q22: Which layer ensures acceptable syntax at both ends of the layer and provides access to adjacent layers.	
A. Network layer	B. Data link layer
C. Session layer	D. Presentation layer
Correct Answer: D	
Reference: Book name IoT NETWORKING, page no 31	

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Q23: Arrange the following components of a fiber-optic cable starting from the center of its cross section; 1:jacket 2:cladding 3:plastic and glass strands 4:buffer	
A. 2,3,1,4	B. 3,2,4,1
C. 3,4,2,1	D. 3,1,2,4
Correct Answer: B	
Reference: Book name IoT NETWORKING, page no 27	

Q24: Which of the following contribute to high power consumptions of end-devices in NB-IoT compared to SigFox and LoRa? [Multiple Choice]	
A. Low latency	B. Synchronous communication
C. QoS	D. FDMA access mode
Correct Answer: B,C,D	
Reference: Book name IoT NETWORKING, page no 76	

Q25: _____ is recommended for IoT connectivity in agricultural farm.	
A. Cellular	B. LPWAN
C. NB-IoT	D. SigFox
Correct Answer: B	
Reference: Book name IoT NETWORKING, page no 81	

Q26: How many devices can be connected per cell in SigFax and LoRa?	
A. 10K	B. 50K
C. 100K	D. 60K
Correct Answer: B	
Reference: Book name IoT NETWORKING, page no 77	

Q27: Which of the following are not include by IIoT?	
A. output	B. customer convenience
C. security	D. ROI
Correct Answer: B	
Reference: Book name IoT NETWORKING, page no 89	

Q28: Which of the following can be used to separate vulnerable smart things from secure devices?	
A. VLANS	B. VPN
C. firewall	D. antivirus
Correct Answer: A	
Reference: Book name IoT NETWORKING, page no 93	

Q29: Which of the following is not a security tool for IoT devices?	
A. Firewall	B. Wi-Fi direct
C. antivirus	D. VLAN
Correct Answer: B	

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Reference: Book name IoT NETWORKING, page no 92

Q30: Which of the following is not a characteristics of smart farming?

- | | |
|-----------------------------|-----------------------------|
| A. Increased waste | B. Optimized fertilizer use |
| C. Optimized vehicle routes | D. Enhanced productivity |

Correct Answer: A

Reference: Book name IoT NETWORKING, page no 93

Q31: Which of the following is used to rework and help in failure management in industries?

- | | |
|-----------------------|----------------------------|
| A. Workshop mirroring | B. Self-dependent system |
| C. Digital twins | D. Supply chain management |

Correct Answer: C

Reference: Book name IoT NETWORKING, page no 95, Heading 6.6

Q32: Which layer has full-duplex and half-duplex modes of operations?

- | | |
|-----------------------|--------------------|
| A. Data link layer | B. Session layer |
| C. Presentation layer | D. Transport layer |

Correct Answer: B

Reference: Book name IoT NETWORKING, page no 31

Q33: Which of the following is immune to EMI and radio frequency interference?

- | | |
|--------|-------------------|
| A. MMF | B. UTP |
| C. STP | D. Co-axial cable |

Correct Answer: A

Reference: Book name IoT NETWORKING, page no 28

Q34: Match the following cable name vs description.

Sl no.	Left column	Sl no.	Right column	Correct answer for column A
A1	APPLICATION LAYER	B1	ENABLES SYNCHRONIZATION OF COMMUNICATIONG NODES	☐ B1 ☒ B2 ☐ B3 ☐ B4
A2	TRANSPORT LAYER	B2	PROVIDES DIRECT ACCESS TO OSI ENVIRONMENT	☐ B1 ☐ B2 ☒ B3 ☐ B4
A3	DATA LINK LAYER	B3	IMPLIMENTS END TO END PROTOCOLS BETWEEN COMMUNICATING NODES	☐ B1 ☐ B2 ☐ B3 ☒ B4

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A4	SESSION LAYER	B4	HAS CONNECTION-ORIENTED AND CONNECTION-LESS OPERATIONG MODES	☒ B1 ☐ B2 ☐ B3 ☐ B4
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Reference: Book name IoT NETWORKING, page no 30-31

Q35: Match the following cable name vs description.

Sl no.	Left column	Si no.	Right column	Correct answer for column A
A1	Fiber optic cable	B1	Ethernet cable	☐ B1 ☐ B2 ☐ B3 ☒ B4
A2	Twisted-pair cable	B2	Provides scope for further development	☐ B1 ☒ B2 ☐ B3 ☐ B4
A3	Co-axail cable	B3	Includes conductor, insulator, braiding and sheath	☐ B1 ☐ B2 ☒ B3 ☐ B4
A4	Cat6a cable	B4	Transmit data up to 40 kilo meters	☒ B1 ☐ B2 ☐ B3 ☐ B4

Reference: Book name IoT NETWORKING, page no 25- 28